VirtualWorks Pharmacovigilance Internship

Task 3 – Causality Assessment

Prepared by: _____
Method: WHO-UMC causality assessment
Document type: Educational simulated case analysis

1. Objective

To evaluate the relationship between a suspected medicine and an adverse reaction using the WHO-UMC causality assessment approach and classify cases as **Certain**, **Probable/Likely**, or **Possible**.

2. WHO-UMC Criteria Used

Category	Key assessment points
Certain	Plausible time relationship; cannot be explained by disease or other drugs; clinically plausible response to withdrawal; clear re-challenge info if available.
Probable / Likely	Reasonable time relationship; unlikely to be explained by disease or other drugs; clinically reasonable improvement after withdrawal; re-challenge not required.
Possible	Reasonable time relationship, but the event can also be explained by disease or another drug; dechallenge information missing or unclear.

3. Case Study 1 – Classified as CERTAIN

Case ID: SIM-CAUS-001
Patient: 28-year-old female
Suspected drug: Ceftriaxone injection
Indication: Bacterial infection
Reaction: Anaphylaxis (hypotension, wheezing and facial swelling)
Timeline: Symptoms began within minutes of administration. The patient had received ceftriaxone previously.
Action/outcome: Ceftriaxone was stopped and emergency treatment was given; symptoms improved.

Assessment: The onset is highly plausible and immediate. The clinical picture is a specific serious allergic phenomenon with a strong temporal relationship. No alternative cause was identified. Rechallenge was not attempted because deliberate re-exposure after suspected anaphylaxis would be inappropriate.

Final classification: CERTAIN.

4. Case Study 2 – Classified as PROBABLE / LIKELY

Case ID: SIM-CAUS-002
Patient: 45-year-old male
Suspected drug: Amoxicillin 500 mg orally three times daily
Indication: Bacterial throat infection
Reaction: Generalized diarrhea
Timeline: Diarrhea started about 2 days after amoxicillin was initiated.
Action/outcome: Amoxicillin was discontinued after clinical review; diarrhea improved over the next few days. No other new medicines were identified and no evidence of a gastrointestinal infection was reported.

Assessment: There is a reasonable temporal relationship. The reaction is a known type of adverse effect associated with antibiotics. Alternative explanations were not strongly supported, and improvement after withdrawal was clinically reasonable. Rechallenge was not required.

Final classification: PROBABLE / LIKELY.

5. Case Study 3 – Classified as POSSIBLE

Case ID: SIM-CAUS-003
Patient: 60-year-old female
Suspected drug: Metformin 500 mg twice daily
Indication: Type 2 diabetes mellitus
Reaction: Dizziness
Timeline: Dizziness was reported after metformin was started. The patient was also taking an antihypertensive medicine and had variable blood-pressure readings.
Action/outcome: No clear dechallenge information was available.

Assessment: The timing provides a reasonable possibility of a relationship with metformin, but dizziness could also be explained by the concomitant antihypertensive treatment or blood-pressure changes. Because alternative explanations are plausible and dechallenge is unclear, the evidence is insufficient for Probable/Likely.

Final classification: POSSIBLE.

6. Summary Table

Case	Drug	Reaction	Classification	Main reason
SIM-CAUS-001	Ceftriaxone	Anaphylaxis	Certain	Immediate, specific clinical phenomenon; strong timing; no alternative cause.
SIM-CAUS-002	Amoxicillin	Diarrhea	Probable / Likely	Reasonable timing, alternative causes unlikely, improvement after withdrawal.
SIM-CAUS-003	Metformin	Dizziness	Possible	Reasonable timing, but other plausible causes exist and dechallenge is unclear.

7. Conclusion

Causality assessment is a structured clinical judgement rather than proof of causation. Timing, alternative causes, dechallenge, clinical characteristics and the quality of available information should be considered. These simulated cases demonstrate how the WHO-UMC approach can lead to Certain, Probable and Possible classifications.

8. Originality and Educational Note

These are original simulated educational cases prepared for internship practice. They are not real patient records.

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